
DETERMINATION OF OPTIMAL LOADING DURING THE POWER CLEAN, IN COLLEGIATE ATHLETES

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ABSTRACT

Comfort, P, Fletcher, C, and McMahon, JJ. Determination of optimal loading during the power clean, in collegiate athletes. *J Strength Cond Res* 26(11): 2970–2974, 2012—Although previous research has been performed in similar areas of study, the optimal load for the development of peak power during training remains controversial, and this has yet to be established in collegiate level athletes. The purpose of this study was to determine the optimal load to achieve peak power output during the power clean in collegiate athletes. Nineteen male collegiate athletes (age 21.5 ± 1.4 years; height 173.86 ± 7.98 cm; body mass 78.85 ± 8.67 kg) performed 3 repetitions of power cleans, while standing on a force platform, using loads of 30, 40, 50, 60, 70, and 80% of their predetermined 1-repetition maximum (1RM) power clean, in a randomized, counter-balanced order. Peak power output occurred at 70% 1RM ($2,951.7 \pm 931.71$ W), which was significantly greater than the 30% ($2,149.5 \pm 406.98$ W, $p = 0.007$), 40% ($2,201.0 \pm 438.82$ W, $p = 0.04$), and 50% ($2,231.1 \pm 501.09$ W, $p = 0.05$) conditions, although not significantly different when compared with the 60 and 80% 1RM loads. In addition, force increased with an increase in load, with peak force occurring at 80% 1RM ($1,939.1 \pm 320.97$ N), which was significantly greater ($p < 0.001$) than the 30, 40, 50, and 60% 1RM loads but not significantly greater ($p > 0.05$) than the 70% 1RM load ($1,921.2 \pm 345.16$ N). In contrast, there was no significant difference ($p > 0.05$) in rate of force development across loads. When training to maximize force and power, it may be advantageous to use loads equivalent to 60–80% of the 1RM, in collegiate level athletes.

KEY WORDS peak power, peak force, rate of force development

INTRODUCTION

Power can be expressed as the product of force and velocity (18), with the highest power during a movement, peak power, being achieved while neither force nor velocity are at their peak. Muscular power is considered one of the main determinants of dynamic athletic performance, especially in sporting events that require high force generation in a short amount of time (18). Training power could therefore have important implications for improving peak power output and have great effects on sport performance. Generally, weight lifting movements and their derivatives are considered highly specific to actual sports performance, because they involve large muscle mass, multijoint movements, and fast movement velocity (2). Such exercises have been suggested to increase an athlete's performance by imitating sport-specific movements, while concurrently using explosive power (22,23), with performance in the hang power clean being correlated to both 20-m sprint and countermovement jump performance (13).

The load at which peak power is produced in lower-body exercises, such as the squat and jump squat, has been reported to vary from 0% (body mass [BM] with no external load) to 60% of 1-repetition maximum (1RM) back squat (3,7,8,17,20,21,23). In contrast to the squat jump, optimal loading during variations of the clean tend to occur between 60 and 80% 1RM power clean (7,11,14–16). Haff et al. (11) found that peak power in the hang power clean also occurred at 80% 1RM, but this was not significantly different from 90 and 100% 1RM; however, testing was not conducted at loads <80% 1RM. Kawamori et al. (15) found that peak power output is achieved at 60% of 1RM (power clean) during the midhigh clean pull, when compared with 30, 90, and 120% of the 1RM power clean. Previously, Kawamori et al. (14) found that peak power output during the hang power clean is achieved using a load of 70% of 1RM power clean. More recently, however, Kilduff et al. (16) found that peak power output during the hang power clean was not significantly ($p > 0.05$) different between loads of 50, 60, 70, 80, or 90% of the 1RM power clean.

It is clear that differing results have been reported, and there is no set agreement among researchers, which may be attributed to technical proficiency of the subjects or methodological issues relating to assessing power during

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variations of the clean. Such large disparity in the research reported has led to ambiguity surrounding the load power relationship (7,8,10). Training with the optimal load is suggested to be the most effective method for improving maximal power and is likely to result in enhancement of a variety of dynamic athletic performances (27). The aim of the study, therefore, was to determine the optimal load at which peak power is achieved during the power clean, in collegiate level athletes, as previous research has only established the optimal load in well-trained professional athletes. It was hypothesized that the optimal load for peak power output, during the power clean, would be achieved at a load of 70% of 1RM power clean, which is in line with the range identified in previous research, using well-trained athletes.

METHODS

Experimental Approach to the Problem

This study employed a within-subjects repeated measures research design, whereby peak power output was determined during the power clean performed at a variety of loads in a randomized counterbalanced order (30, 40, 50, 60, 70, and 80% 1RM power clean) to determine which relative load results in the greatest power output. Dependent variables, peak vertical ground reaction force (F_z), peak rate of force development (RFD), and peak power were measured while the athletes performed all exercise variations while standing on a force platform (Kistler, Winterthur, Switzerland, Model 9286AA, SN 1209740). These kinetic variables were selected as F_z and measures such as RFD have been shown to be strong determinants of sprint performance (24–26).

Subjects

Nineteen healthy male collegiate athletes (age 21.5 ± 1.4 years; height 173.86 ± 7.98 cm; BM 78.85 ± 8.67 kg; 1RM power clean 84.52 ± 7.35 kg) participated in this study. All the participants had regularly ($>3 \times$ week) performed structured strength and conditioning training in preparation for their sport (rugby, field hockey, soccer), including variations of the clean, for >1 year. The investigation was approved by the Institutional Ethics Review Board, and all the subjects provided informed consent before participation. The study conformed to the principles of the World Medical Association's Declaration of Helsinki. The participants had previously conducted technique sessions, supervised by a certified strength and conditioning coach, within their normal training to allow familiarization with the protocols and ensure appropriate technique. Testing took place during the competitive season, after the participants had completed a power mesocycle.

Testing

The 1RM power cleans were assessed on 2 separate occasions, at the same time of the day, 3–5 days apart, to determine reliability following a standardized protocol (1). The subjects were asked to replicate their fluid and food intake on both days and avoid strenuous exercise for 24 hours

before testing. After both the 1RM testing sessions, each subject was familiarized with the protocols for the power testing of each exercise. Before power testing, all the subjects performed a standardized dynamic warm-up, including each variation of the power clean (4 repetitions, 3 sets) using a standardized load (30 kg) (Werksan weights and Olympic bar; Werksan, Morristown, NJ, USA). The participants were then randomly assigned to perform 1 cluster set of 3 repetitions (60-second rest between repetitions to minimize fatigue) of the power clean (bar starting midway up the shin and caught in a shallow squat, for each load. Four minutes of rest between each load was provided to ensure adequate recovery time, which is in line with the findings of previous research (7,8,14).

Each repetition was performed with the subjects standing on a force plate, sampling at 1,000 Hz, interfaced with a laptop. Data were later analyzed using Bioware (Version 3.22; Kistler Instrument Corporation) to determine peak F_z . Instantaneous RFD was determined by dividing the difference in consecutive F_z readings by the time interval (0.001 seconds) between readings. Data were smoothed using a moving average window of 400 milliseconds. Velocity of the center of gravity (COG) of the system (barbell + body) was calculated from F_z time data based on the relationship between impulse and momentum in which impulse is equal to the changes in momentum (forward dynamics approach). Lower-body power applied to the system was calculated as the product of velocity of the COG of the system and F_z at each time point

TABLE 1. Intraclass correlation values for mean peak force, mean peak power, and mean peak rate of force development at various loads.*

Load (% 1RM)		<i>r</i> Value	<i>p</i>
30	F_z	0.936	<0.001
	Peak power	0.845	<0.001
	RFD	0.790	<0.001
40	F_z	0.962	<0.001
	Peak power	0.868	0.001
	RFD	0.923	<0.001
50	F_z	0.971	<0.001
	Peak power	0.836	<0.001
	RFD	0.894	<0.001
60	F_z	0.936	<0.001
	Peak power	0.893	0.002
	RFD	0.887	<0.001
70	F_z	0.957	<0.001
	Peak power	0.828	<0.001
	RFD	0.912	<0.001
80	F_z	0.940	<0.001
	Peak power	0.880	0.002
	RFD	0.852	<0.001

*RFD = rate of force development; RM = repetition maximum.

TABLE 2. Mean and SD values for peak force production during the power clean at various loads.*

Load (% 1RM)	Mean (SD) (N)	95% Confidence interval	
		Lower bound	Upper bound
30	1,561.105 (220.18)	1,454.982	1,667.228
40	1,621.184 (249.61)	1,500.875	1,741.493
50	1,695.921 (296.26)	1,553.127	1,838.715
60	1,817.588 (271.98)	1,686.499	1,948.677
70	1,921.245 (345.16)	1,754.881	2,087.608
80	1,939.167 (320.97)	1,784.465	2,093.869

*RM = repetition maximum.

(12). When calculating power using F_z , the impulse-momentum approach is used to calculate power, where impulse is equal to a change in momentum, or force multiplied by time. Because the force, system mass, and initial velocity conditions are known, the instantaneous velocity can be calculated using this approach. Power can then be calculated as force multiplied by velocity, and the peak of these values can be determined for the propulsive phase of each variation of the power clean. For each i , or time point based on sampling frequency (equation set for the force data only):

$$\begin{aligned}
 v_{(0)} &= 0, \\
 F_{(i)}t &= m(v_{(i+1)} - v), \\
 &= (F_{(i)}t)/m, \\
 P_{(i)} &= F_{(i)} \times v_{(i)},
 \end{aligned}$$

where F is the force, t is the $1/\text{sampling frequency}$, m is the mass of body 1 load, v is the velocity, and P is the power.

To implement this calculation method, the sampling rate and F_z are needed, along with an initial velocity of the system of zero. To calculate power in this way, it was important that

the initial F_z represented system load (athlete's BM plus load lifted); consequently, the bar was held slightly above ground level before the onset of the power clean, in line with what was done in previous research (5,6). Power is calculated along the vertical axis only and is the result of lower-body force production and not representative of the power applied to the bar.

Statistical Analyses

Intraclass correlation coefficients (ICCs) were calculated

to determine reliability between 1RM power cleans and to establish reproducibility between repetitions during each exercise variation. A 1-way analysis of variance and Bonferroni post hoc analysis were conducted to determine if there were any significant differences in dependent variables (peak power output, RFD, and F_z) between relative loads. Statistical power was calculated between 0.89 and 0.92 for each loading condition. An a priori alpha level was set to $p \leq 0.05$.

RESULTS

The ICCs show a high reliability for peak F_z ($r > 0.936$, $p < 0.01$) and peak power output ($r > 0.828$, $p < 0.001$), with a moderate to high reliability for RFD ($r > 0.790$, $p < 0.001$) across all loads, in line with the recommendations of Cortina (9) (Table 1).

Force Production

Force production increased as load increased, with the peak F_z produced at 30% ($1,561.1 \pm 220.18$ N, $p < 0.001$), 40% ($1,621.1 \pm 249.61$ N, $p < 0.001$), and 50% ($1,695.9 \pm 296.26$ N, $p < 0.003$) being significantly lower than the 60, 70, and 80% 1RM loading conditions. Peak F_z occurred at 80% 1RM ($1,939.1 \pm 320.97$ N), which was significantly greater ($p < 0.001$) than the 30, 40, 50, and 60% 1RM loads but not significantly greater ($p > 0.05$) than the 70% 1RM load ($1,921.2 \pm 345.16$ N) (Table 2).

Peak Power

Peak power output occurred at 70% 1RM ($2,951.7 \pm 931.71$ W), which was significantly greater than the 30% ($2,149.5 \pm 406.98$ W, $p = 0.007$), 40%

TABLE 3. Mean and SD values for peak power production during the power clean at various loads.*

Load (% 1RM)	Mean (SD) (W)	95% Confidence interval	
		Lower bound	Upper bound
30	2,149.544 (406.98)	1,953.384	2,345.704
40	2,201.009 (438.82)	1,989.500	2,412.571
50	2,231.114 (501.09)	1,989.596	2,472.632
60	2,705.281 (624.47)	2,404.296	3,006.265
70	2,951.702 (931.71)	2,502.631	3,400.774
80	2,918.614 (1022.58)	2,425.744	3,411.483

*RM = repetition maximum.

TABLE 4. Mean and *SD* values for peak rate of force development during the power clean at various loads.*

Load (% 1RM)	Mean (<i>SD</i>) (N·s ⁻¹)	95% Confidence interval	
		Lower bound	Upper bound
30	8,839.912 (3,185.64)	7,304.482	10,375.342
40	8,748.123 (3,328.16)	7,144.000	10,352.245
50	9,288.509 (3,600.49)	7,553.126	11,023.892
60	10,227.227 (3,750.86)	8,419.369	12,035.086
70	10,741.912 (4,291.02)	8,673.709	12,810.115
80	10,700.746 (2,946.02)	9,280.811	12,120.681

*RM = repetition maximum.

(2,201.0 ± 438.82 W, $p = 0.04$), and 50% (2,231.1 ± 501.09 W, $p = 0.05$) 1RM conditions, although not significantly different ($p > 0.05$) than the 60 and 80% 1RM conditions (Table 3).

Rate of Force Development

In general, the peak RFD increased as load increased, with the greatest peak RFD occurring at 70% 1RM (10,741.9 ± 4,291.02 N·s⁻¹); however, this was not significantly different ($p > 0.05$) to the RFD produced with any other load (Table 4).

DISCUSSION

The primary finding from this study was that peak power output (2,951.7 ± 931.71 W) was maximized at 70% 1RM in the power clean, which is in line with the original hypothesis; however, peak power output at 60, 70, and 80% of 1RM were not significantly ($p > 0.05$) different, in line with the findings of previous research using the hang power clean (14). This confirms suggestions that peak power output may be a very individual response and can occur at any of the 3 relative loads of 60, 70, and 80% of 1RM, although Kilduff et al. (16) found that peak power output occurred at 80% 1RM. In fact, individual results in this study show that 5 subjects achieved their peak F_z , RFD, and Power at 60%, 6 at 70%, and 9 at 80%, demonstrating the aforementioned individual response.

The results of this study are also comparable with results found by Haff et al. (11), who reported that peak power output occurred at 80% 1RM (2,440.23 ± 236.90 W); however, they only tested at loads of 80, 90, and 100% of 1RM, and therefore, it cannot be discounted that peak power may have occurred at a load <80% 1RM. Although the peak power output (2,951.7 ± 931.71 W) achieved in this study is similar to the findings of Haff et al. (11) (2,440.23 ± 236.90 W), it was substantially lower than the peak power outputs achieved in the studies of Kilduff et al. (15) (4,460.7 ± 477.2 W) and Kawamori et al. (14) (4,281.15 ± 634.84 W). This may be attributed to the higher BM and absolute strength (BM = 102.4 ± 11.4 kg, 1RM = 107 ± 13 kg;

BM = 89.4 ± 14.7 kg, 1RM = 107.0 ± 18.8 kg, respectively) of the subjects of the later studies compared with this study (BM = 78.85 ± 8.67 kg; 1RM 84.52 ± 7.35 kg). It is suggested, therefore, that collegiate level athletes should perform the power clean with a load of 60–80% 1RM maximize power output, which is in line with previous research using more experienced athletes (4,14–16) and to account for the individual variation noted above.

The F_z increased as load increased, with the greatest

peak F_z (1,939.1 ± 320.97 N), occurring at the highest load (80% 1RM), although this was not significantly different from the peak F_z produced at 70% 1RM (1,921.2 ± 345.16 N), which is in agreement with previous findings (14,16). Individual results also showed some individual variation with peak F_z and RFD occurring between 60 and 80% 1RM, mirroring the individual variations in peak power already discussed. In contrast the higher absolute peak F_z reported by Kilduff et al. (15) ($F_z = 3,487.0 ± 526.6$ N) compared with this study (1,939.1 ± 320.97 N) may be attributable to the lower system mass (BM + bar mass) in this study.

Peak RFD occurred at 70% of the 1RM, although interestingly this was not significantly different from any of the other loads tested, which may be explained by Schmidbleicher (19) who reported the peak RFD was equal for all loads >25% of peak F_z .

It is suggested that further research be conducted to determine whether training at the load that maximizes individual peak power output, compared with training at higher, or lower relative loads, results in a greater adaptive response. It would also be advantageous to see if any improvements in F_z , power, or RFD are related to any subsequent changes in sprint or jump performance.

PRACTICAL APPLICATIONS

The findings of this study indicate that when training to maximize peak power output, a load of 70% 1RM power clean may be advantageous; similarly, if the focus is developing or maintaining peak F_z 80% 1RM may be optimal. It is noteworthy, however, that individual responses to loading varied with peak values occurring between 60 and 80% 1RM across individuals. It is suggested, therefore, that when developing training programs for collegiate athletes which include the power clean, a range of loads, between 60–80% 1RM, and identification of the loads that elicit peak power in individual athletes may be advantageous, because of the individual responses noted.

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